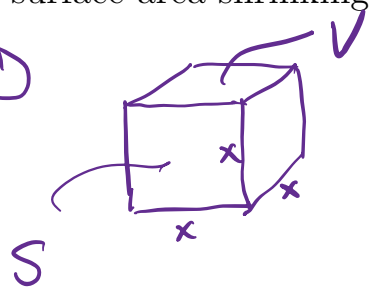
Problem 8. TIER 1 — RUN IT BACKWARDS

A cube of ice melts, keeping its cube shape, losing volume at 2 cm^3/min. When the edge is 4 cm, how fast is the edge shrinking? How fast is the surface area shrinking?

1. Draw + Label
2. What & when
3. Equation
4. Differentiate w.r.t. t
5. Sub + Solve



(2) $\frac{dV}{dt} = -2$

$\frac{dx}{dt} = ?$, $\frac{dS}{dt} = ?$

$x = 4$

(3) $V = x^3$

(4) $\frac{d}{dt}V = \frac{d}{dt}x^3$

$\frac{dV}{dt} = 3x^2 \frac{dx}{dt}$

(5) $-2 = 3 \cdot 4^2 \frac{dx}{dt}$

$\frac{-2}{3 \cdot 16} = \frac{dx}{dt}$

$\frac{dx}{dt} = \frac{-1}{24} \frac{\text{cm}}{\text{min}}$

$S = 6x^2$

$\frac{d}{dt}S = \frac{d}{dt}6x^2$

$\frac{dS}{dt} = 12x \frac{dx}{dt}$

$\frac{dS}{dt} = 12 \cdot 4 \left(\frac{-1}{24}\right)$

$= \frac{48}{24}(-1) = -2 \frac{\text{ft}^2}{\text{min}}$

negative